CCP4 SHANGHAI WORKSHOP

Coot and Moorhen

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IMAGES FROM PAUL EMSLEY

Coot – Crystallographic Object Oriented Toolkit

Version 0.2

Version 0.3

Version 0.4

Version 0.5

Version 0.6

Version 0.7

Version 0.8

Version 0.9

Versions of Coot over the last 22 years

Legacy of coot

Coot has been around for a long time – over 20 years

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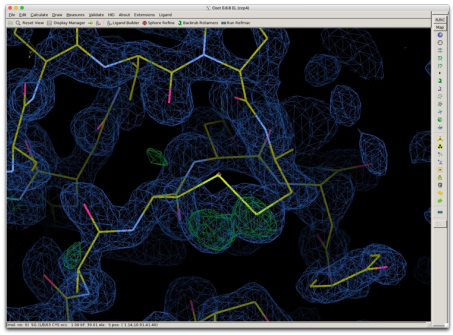
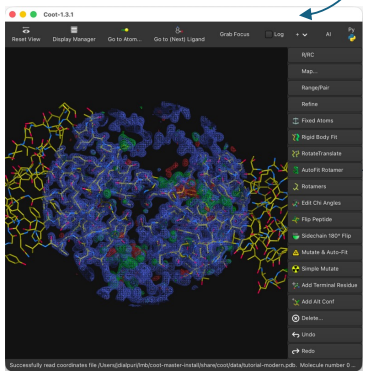
Two versions of Coot - which is which?

Recommended



Coot 0.X



Coot 1.X

Two versions of Coot, distinguishable by colour

Two version of Coot which are used today, Coot 0.X which you can tell based on the grey background, and Coot 1.X which is shown by the black background. Coot 0.X is no longer developed and it is recommended to use Coot 1.1

Model building with Coot

`Tools to help improve the quality of the macromolecular model`

Mutating residues

Adding or deleting residues

Merging residues into chains

Refine residues

Validate models

Fit models

Some of the things coot can do

You let automated tools do what they can – and now what? This is where you would use coot
What we want to do when using coot is find problems in our model and fix them, coot provides lots of tools to do this

IMAGES FROM PAUL EMSLEY

Can you do manual model building with Coot?

Manual model-building involves creating a physical model

Very **time consuming** and physically **intensive**

What does Coot do?

Coot is **fast** and **automated**

You can do **interactive model building** with Coot

Designed to **fill the gap** where automated methods fail

Dorothy Hodgkin building the first insulin model

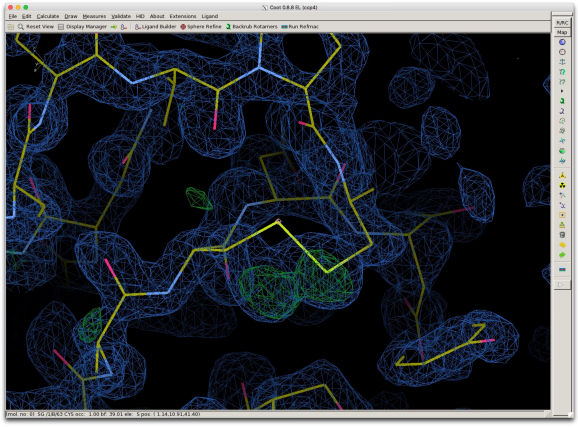
William Lipscomb modelling carboxipeptidase

Photographs of manual model building

Coot is not manual model building, Coot allows us to do interactive model building

IMAGES FROM PAUL EMSLEY

You can use Coot to fix what automated methods get wrong

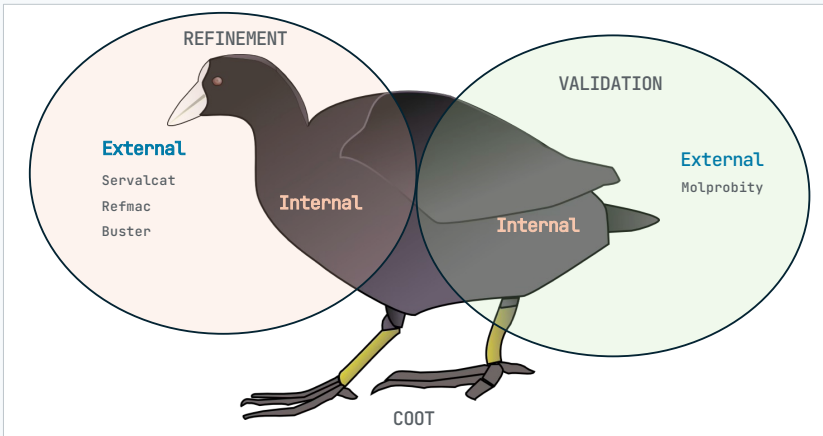


Disulphide bridge mismodelled

Typical example where AMB builds it one way (wrong), we know it's wrong because we see the model, 2mfo-dfc and anomalous map
Clarifies our understanding of where the atoms are

IMAGES FROM PAUL EMSLEY

You can use Coot to fix what automated methods leave unfinished

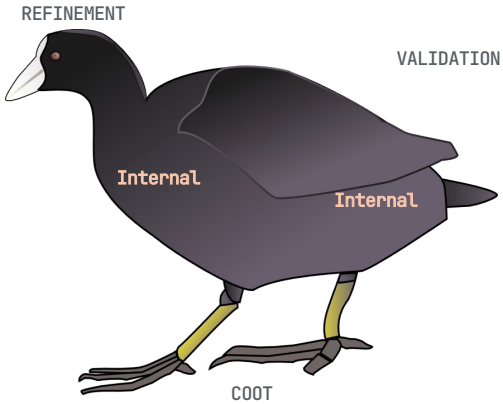


Coot is the connector between other software packages

Coot is an env where we can do all these things – the fact that they are linked makes it easy

IMAGES FROM PAUL EMSLEY

You can use Coot to fix what automated methods leave unfinished



Coot is the connector between other software packages

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

Major feature of Coot

Gradient-based minimiser (BFGS derivative)

Provides **interactive, real space** refinement

Minimises **bond length, angles, torsions, planes, chiral volume and non-bonded contact**

Works with **links** and **modifications**

Been updated many times over the years
Now is detached out of main thread so doesn't stop UI
Multithreaded – semaphores

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

How does it work?

Relies on our **understanding of chemistry**

We know **bond lengths** and uncertainties

We know **bond angles** and uncertainties

We know **the chiral centers**

We know which atoms should lie in a **plane**

We know about **torsions** (mostly)

We can combine the gradients from **the data** with those from **the molecular mechanics** during the minimisation

```
data_comp_list
loop_
_chem_comp.id
_chem_comp.three_letter_code
_chem_comp.name
_chem_comp.group
_chem_comp.number_atoms_all
_chem_comp.number_atoms_nh
_chem_comp.desc_level
ATP     ATP      "ADENOSINE-5'-TRIPHOSPHATE"     NON-POLYMER     43     31     .
loop_
_chem_comp_bond.comp_id
_chem_comp_bond.atom_id_1
_chem_comp_bond.atom_id_2
_chem_comp_bond.type
_chem_comp_bond.aromatic
_chem_comp_bond.value_dist_nucleus
_chem_comp_bond.value_dist_nucleus_esd
_chem_comp_bond.value_dist
_chem_comp_bond.value_dist_esd
ATP       "C5'"      "H5'1"      SINGLE       n     1.087  0.0100     0.989  0.0200
ATP       "C5'"      "H5'2"      SINGLE       n     1.087  0.0100     0.989  0.0200
ATP       "C4'"       "H4'"      SINGLE       n     1.087  0.0100     0.981  0.0200
ATP       "C3'"       "H3'"      SINGLE       n     1.087  0.0100     0.992  0.0200
ATP       "O3'"      "HO3'"      SINGLE       n     0.969  0.0180     0.849  0.0200
ATP       "C2'"       "H2'"      SINGLE       n     1.087  0.0100     0.994  0.0200
ATP       "O2'"      "HO2'"      SINGLE       n     0.969  0.0180     0.849  0.0200
ATP       "C1'"       "H1'"      SINGLE       n     1.087  0.0100     0.984  0.0200
ATP          C8          H8      SINGLE       n     1.083  0.0150     0.942  0.0170
ATP          N6        HN61      SINGLE       n     1.014  0.0120     0.877  0.0200
ATP          N6        HN62      SINGLE       n     1.014  0.0120     0.877  0.0200
ATP          C2          H2      SINGLE       n     1.083  0.0150     0.945  0.0200
```

chem_comp_block entry for ATP from CCP4 Monomer Library

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

How does it work?

We can combine the gradients from **the data** with those from **the molecular mechanics** during the minimisation

The function we are trying to minimise is *S*, where:

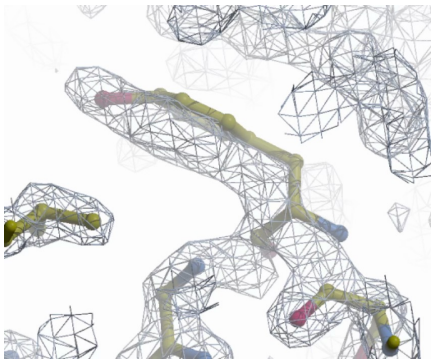
$$S = S_{bond} + S_{angle} + S_{torsion} + S_{plane} - S_{density}$$

Try to minimise the deviation from ideals and fit to density (highly positive where there is most electrons)
- fits to positive density

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?



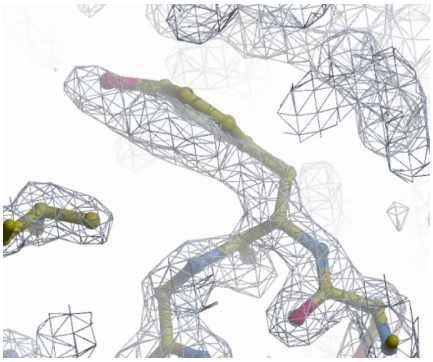
Amino acid slightly misplaced

Tyr out of density and not planar

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?

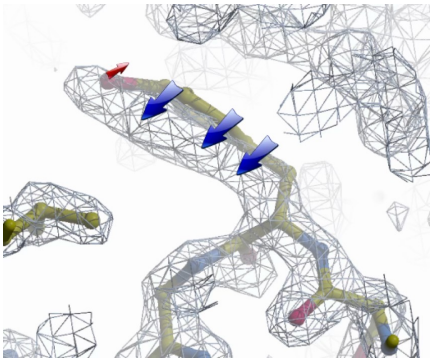


Backbone bond lengths and angles resolved

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?



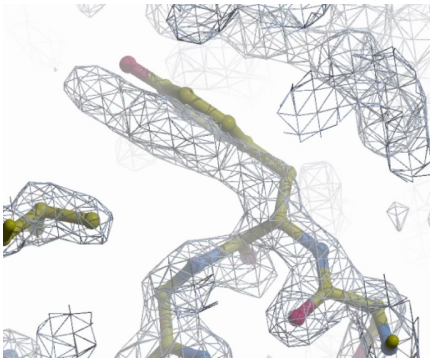
Planarity of tyrosine incorrect and out of density

Blue is density
Red is geometry

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?

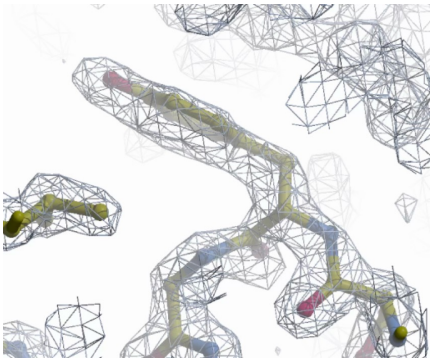


Planarity of tyrosine resolved

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?

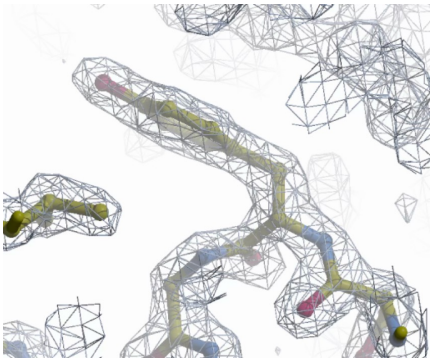


Sidechain moved into density

IMAGES FROM PAUL EMSLEY

Real space refinement in Coot

What does it look like?



Sidechain moved into density

Consistent w our prior knowledge, bond lengths angles and planarity and density

IMAGES FROM PAUL EMSLEY

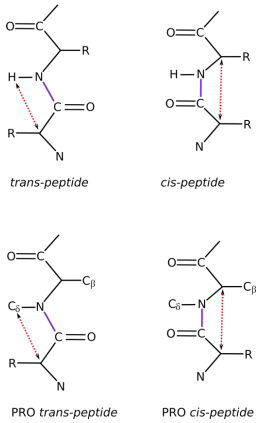
Coot can help spot and fix cis-peptides

What is a cis-peptide?

A number of papers have been published which highlight an unusually large number of cis-peptides in some structures:

Croll, The rate of cis-trans conformation errors is increasing in low-resolution crystal structures Acta Cryst. (2015). D71, 706-709

Touw et al, The rate of cis-trans conformation errors is increasing in low-resolution crystal structures Acta Cryst. (2015). D71, 706-709

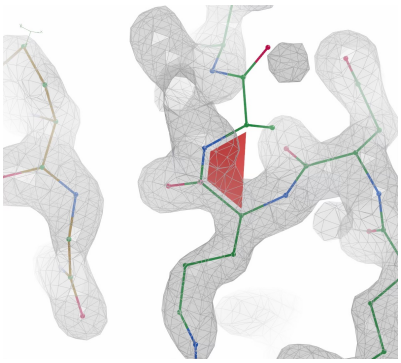


Trans and cis peptides

Early days – there were not restraints which stopped trans/cis peptides
Often people moved structure around and as they did so the backbone was distorted
Moving the N over to the same side as the C=O, the planar restraints kept it in cis
Coot showed cis/trans warning during RSR

IMAGES FROM PAUL EMSLEY

Coot can help spot and fix cis-peptides

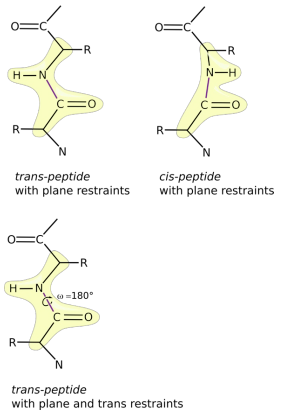


Coot highlights cis-peptides

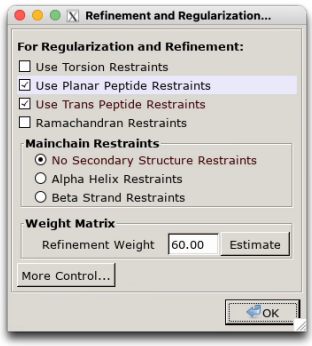
Now shows this to warn you

IMAGES FROM PAUL EMSLEY

Coot can help spot and fix cis-peptides



Plane and trans restraints help mitigate these issues



Tick boxes used to enable these in refinement

Trans and planar restraints in Coot help minimise occurrence of cis peptides.

Now there are trans restraints if the peptide was trans before

IMAGES FROM PAUL EMSLEY

Coot can identify unlikely rotamers and fix them

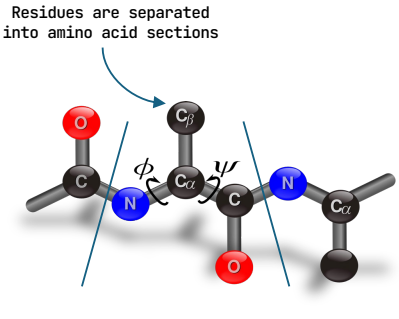
What is a rotamer?

Rotamers are preferred configurations of the **rotatable bonds** of **side-chains**

Preferred means that these configurations occur more frequently in a set of reference protein structures

Preferred because they are low-energy confirmations

Many rotamer libraries exist, Coot uses a version of the *Penultimate Rotamer Library*

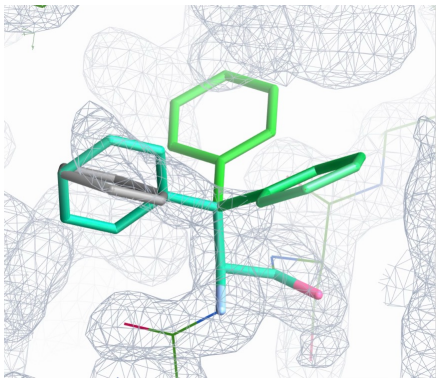


Peptide backbone geometry

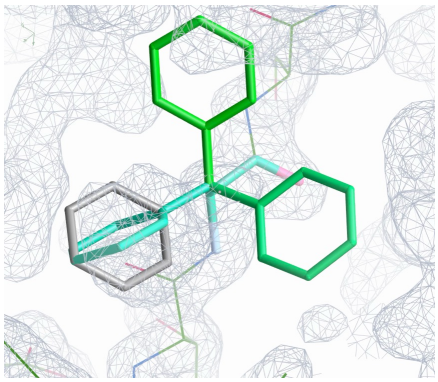
Point of this slide is to show that the N-CA-C-O is a unit – this is 1 entity
Coot would only move this residue but peptide is connected which was a bad approach – specifically at low resolution when the density is unclear

IMAGES FROM PAUL EMSLEY

An example – four rotamers of phenylalanine



Side view



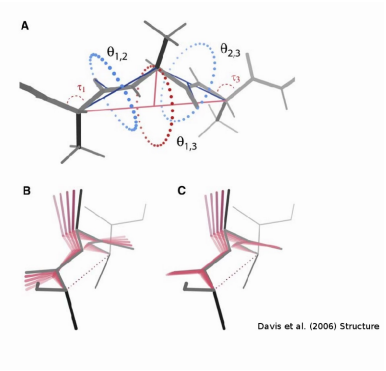
Top view

Four rotamers of phenylalanine that can be trialled with Coot.

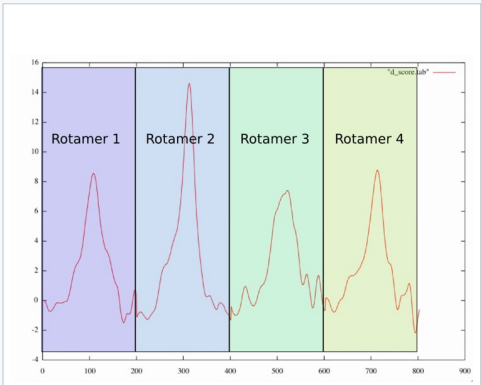
4 rotamer w > 1% probability
Most PHE are in one of these 4
So all we need to do is check all 4 chi angles to the density

IMAGES FROM PAUL EMSLEY

Sometimes, no rotamer fits the density well because of the backbone



Backrub rotamer algorithm



Rotamer scoring per backbone rotation

Backrub rotamer algorithm in Coot moves the backbone and trials various rotamers to get the best fit.

Extra dimension of rotamer search

Draw line between previous and next c alpha - the moving atom set is all atoms in between
Brings along some of the atoms of the neighbouring residues

1st move around red circle (-30 +30), for each orientation on red, you try all rotamers
And then can compensate for distorted backbone geometry with blue circle movement

Fit to density measures + clash check
Deletes water if there any clashes

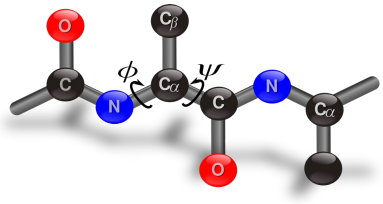
IMAGES FROM PAUL EMSLEY

Coot can show Ramachandran plots

What is a Ramachandran plot?

The backbone of a protein can be defined in torsion angles ϕ and ψ

A Ramachandran plot **shows allowed and disallowed regions** graphically



Peptide backbone geometry

IMAGES FROM A. THORN AND B. LOHKAMP

Coot can show Ramachandran plots

What do they look like in Coot?

Ramachandran plots shown in Coot depend on what sidechain is highlighted.

Describe why they are different
Different residues have different types of plot because they have a different geometry

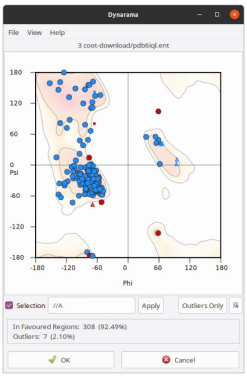
GLY / Pre-PRO / PRO are separated because they are different
ILE/LEU is slightly different distribution
General is everything else

All is shown if you see all residues

IMAGES FROM A. THORN AND B. LOHKAMP

Coot can show Ramachandran plots

What do they look like in Coot?

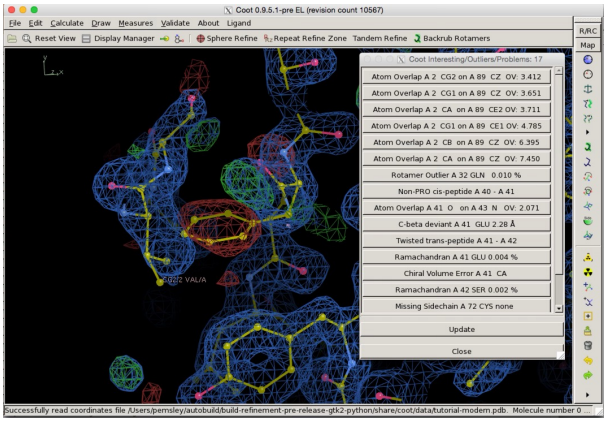


Interactive panel in Coot

Red dots are clickable

IMAGES FROM PAUL EMSLEY

Coot provides interactive validation, allowing interactive checking and fixing

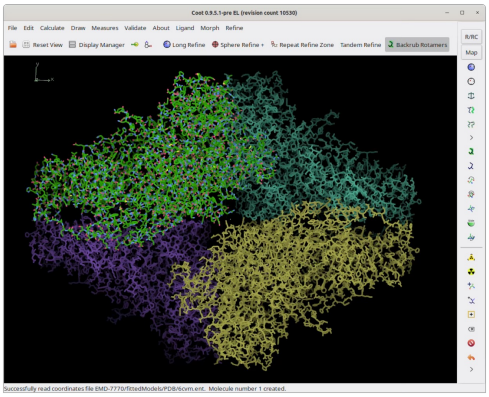


Interactive validation panel

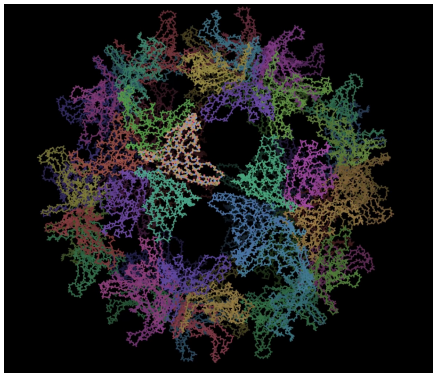
You can click on the boxes to go to the problem

IMAGES FROM PAUL EMSLEY

Coot can show crystallographic symmetry copies



Beta-galactosidase D2 symmetry



Adenovirus symmetry

Coot shows symmetry copies with bonds all of the same colour.

As you build a molecule, using molecular symmetry you don’t build into sym copies

IMAGES FROM TRISTAN CROLL

Coot can update electron density maps on the fly

How does it do this?

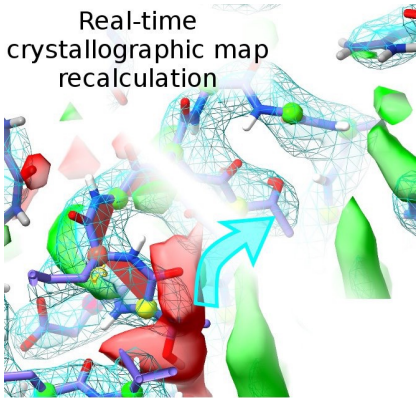
Coot recalculates the electron density map when a model **has been edited**

Using the most up to date model, a **new set of phases** can be estimated

Inspired by Tristan Croll's **ISOLDE**

Implementation aided by Kathryn Cowtan

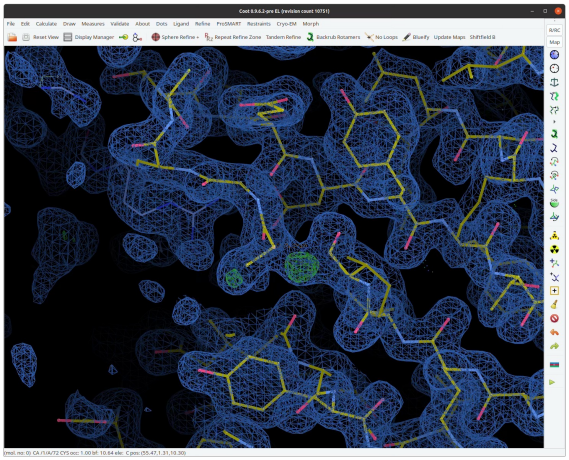
Takes **~0.7s** for a model with 200 amino acids at 1.8 Å resolution



Map recalculation in ISOLDE

IMAGES FROM PAUL EMSLEY

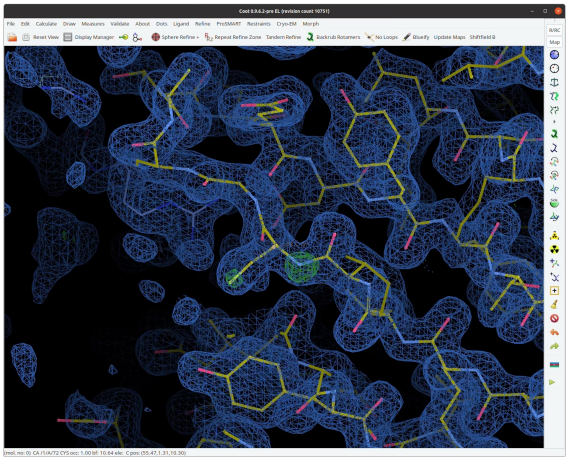
Map recalculation in Coot



Starting point

IMAGES FROM PAUL EMSLEY

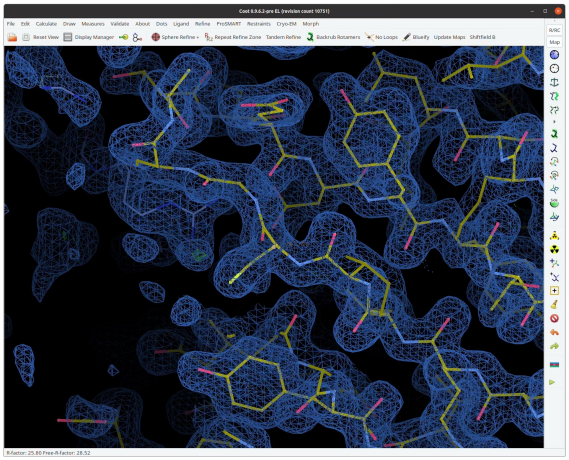
Map recalculation in Coot



An edit to the model

IMAGES FROM PAUL EMSLEY

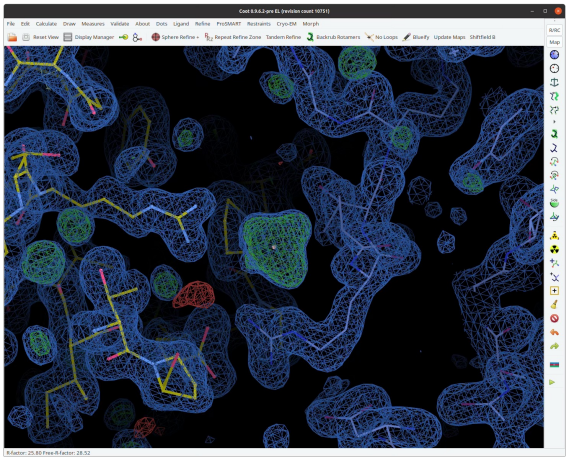
Map recalculation in Coot



Map updated

IMAGES FROM PAUL EMSLEY

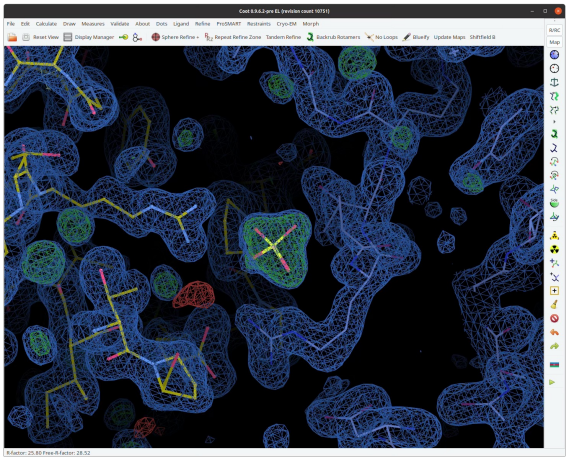
Map recalculation in Coot



Another starting point

IMAGES FROM PAUL EMSLEY

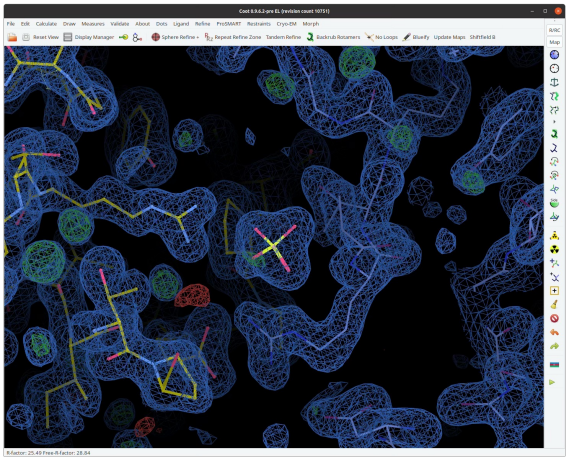
Map recalculation in Coot



Sulphate ion placed

IMAGES FROM PAUL EMSLEY

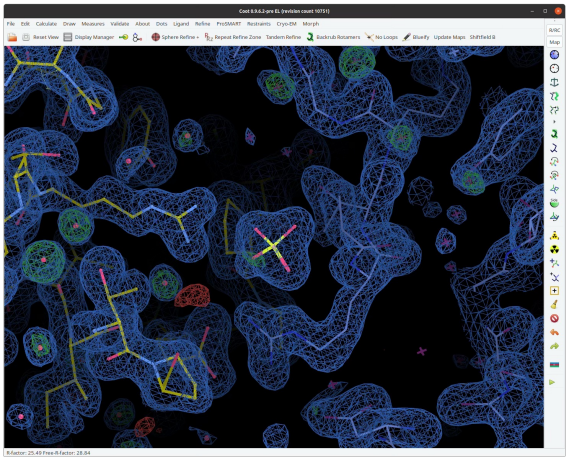
Map recalculation in Coot



Map recalculated

IMAGES FROM PAUL EMSLEY

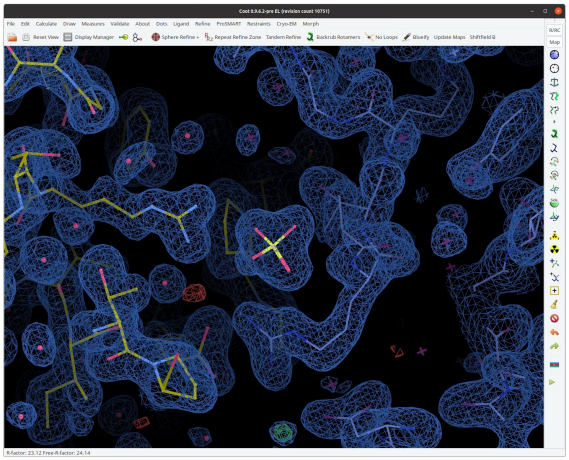
Map recalculation in Coot



Waters added

IMAGES FROM PAUL EMSLEY

Map recalculation in Coot



Map recalculated

Coot 1.1

Coot 1.1 is a major change from Coot 0.X

What's changed?

All features are in a **single panel view**

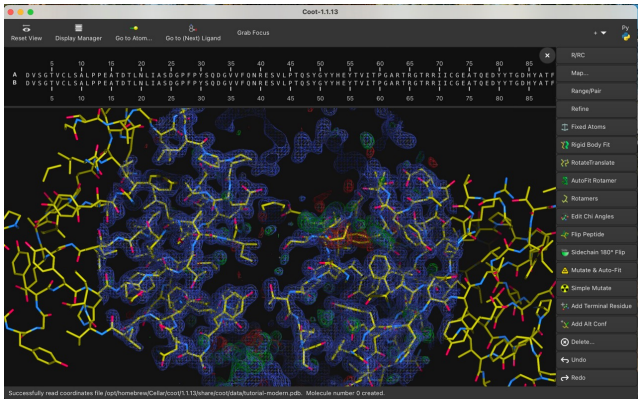
- Sequence View
- Ramachandran plot
- Geometry analysis

Lighting and **representations** have been improved

Real space refinement interface improved

IMAGES FROM PAUL EMSLEY

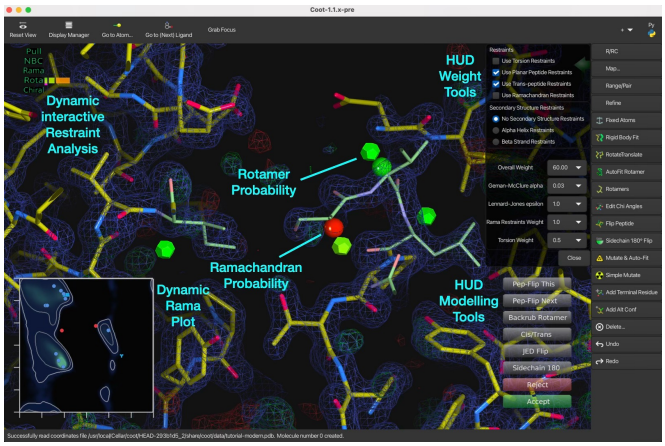
Sequence view is now internal to Coot



Sequence view is embedded in the main Coot window

IMAGES FROM PAUL EMSLEY

Improved interface for interactive real space refinement



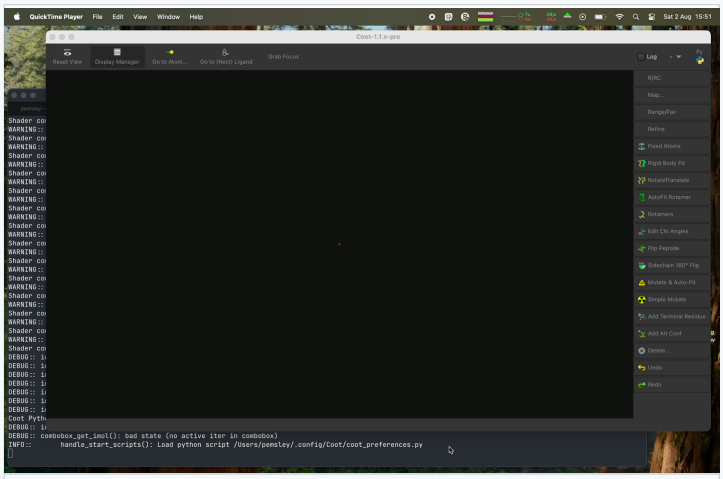
More required information available on the screen

Modern interface to refinement, interactive validation

Bars
Every restraint is analysed every 1/10th second, the whole list of restraints is analysed and the ones which don't get fixed with the mimimiser are highlighted
The colour of the block is a function of the value getting added by the target function
Width is similar

IMAGES FROM PAUL EMSLEY

Improved interface for interactive real space refinement

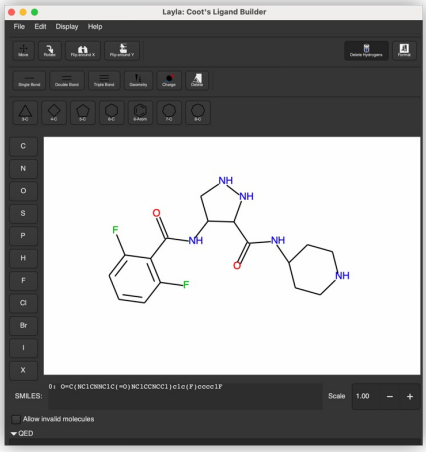


Demonstration of interactive real space refinement

Can turn rotation between left and right
Validation
Different types of validation
Rsr on
Circles rama
Dodec rota

WORK FROM JAKUB SMULSKI – GLOBAL PHASING LIMITED

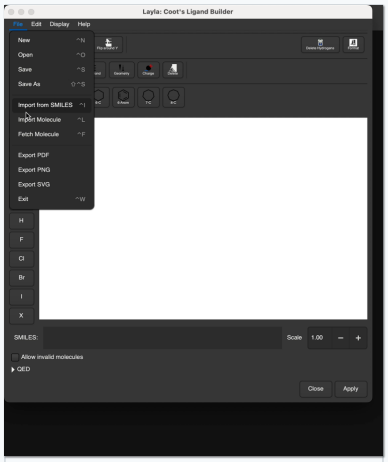
Coot 1.1 has a new ligand editor



New editor - Layla

WORK FROM JAKUB SMULSKI – GLOBAL PHASING LIMITED

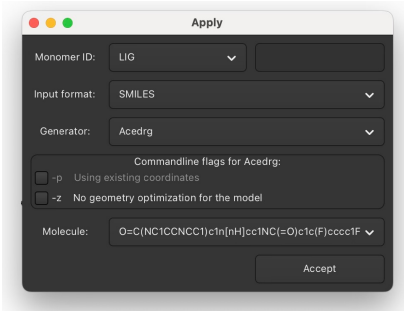
Layla can help guide ligand design



Demonstration of drawing within Layla

WORK FROM JAKUB SMULSKI – GLOBAL PHASING LIMITED

Layla can connect to AceDRG to generate restraints



AceDRG dialogue within Layla

Coot 1.1 can make publication quality figures

IMAGES FROM PAUL EMSLEY

Coot 1.1 can produce beautiful graphics

Possible visual representations of a molecule in Coot 1.1

Changing from coot being tool from model editing – and now how to make pretty pictures

IMAGES FROM PAUL EMSLEY

Coot 1.1 has a ‘fancy’ mode

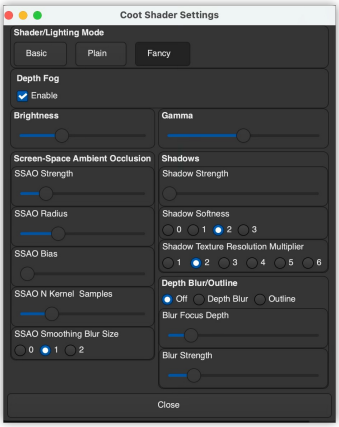
What can it do?

Frame-buffer effects

Screen Space Ambient Occlusion (SSAO)

Shadows

Outlines and Blur

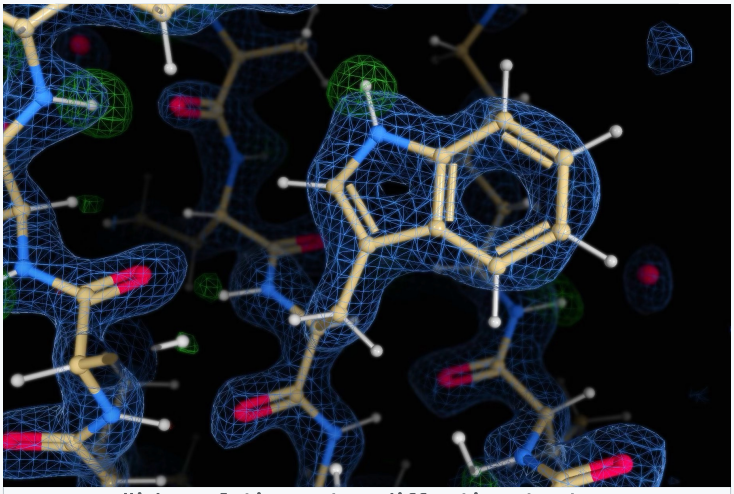


Coot shader settings

SSAO puts shadows into cavities and makes it look photoreal

IMAGES FROM PAUL EMSLEY

Fancy graphics



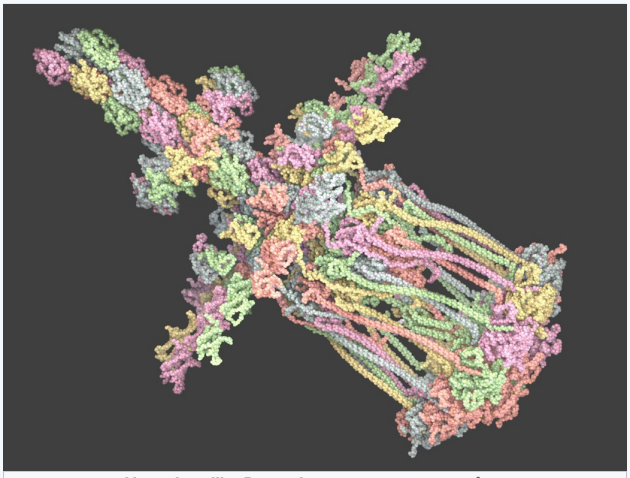
High resolution neutron diffraction structure

H res neutron structure
Depth of field blurring
Mild projection matrix – rather than orthographic

GAMES use SSAO
ChimeraX precalculated AO because

IMAGES FROM LUCREZIA CATAPANO

Fancy graphics

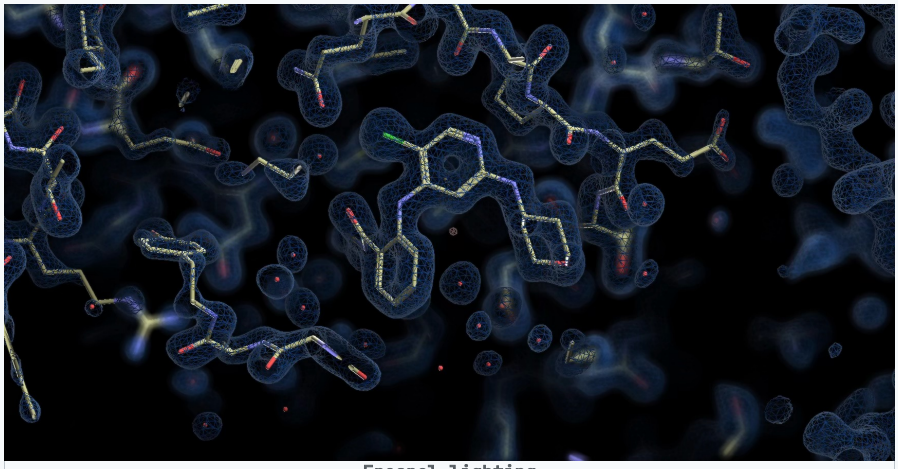


Van der Waals sphere representation

Big molecules
Colour by chain
Coot.set_model_molecule_representation_style(1) gives vdW spheres

IMAGES FROM PAUL EMSLEY

Fancy graphics

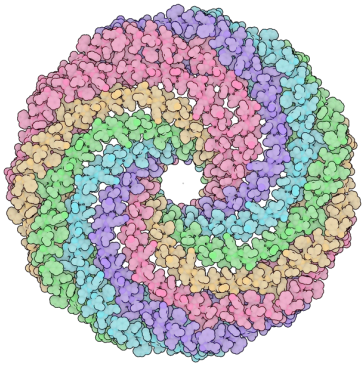


Fresnel lighting

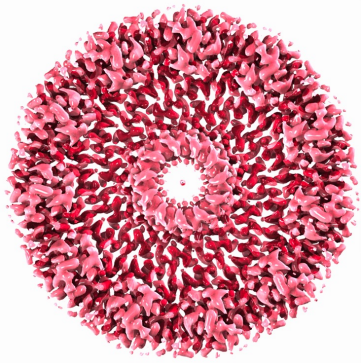
Fresnel lighting is an approximation
Rim lighting

IMAGES FROM LUCREZIA CATAPANO

Fancy graphics



Goodsell style

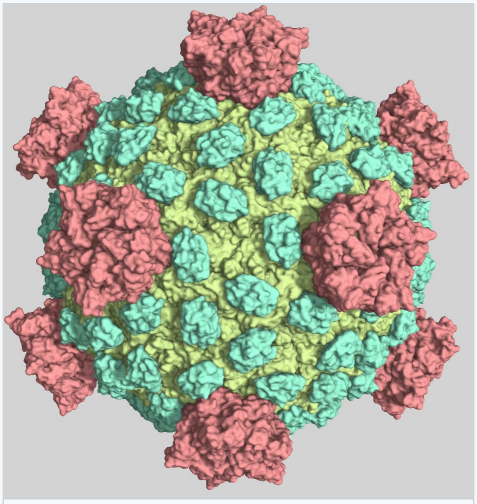


Cryo-EM isosurface

Goodsell colour by chain

IMAGES FROM PAUL EMSLEY

Coot 1.1 can produce Gaussian surfaces

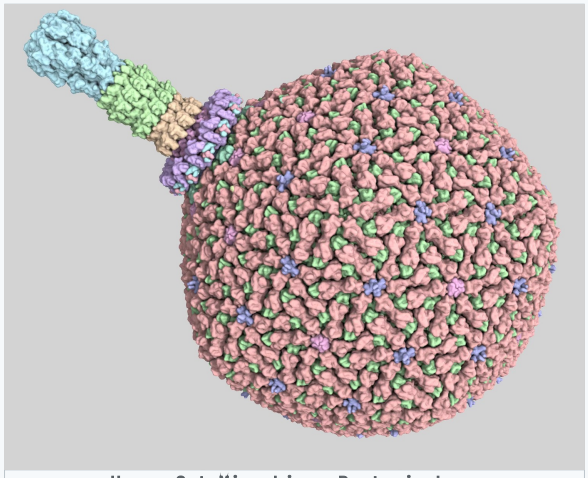


Gaussian surface of Reovirus core

1ej6
With molecular symmetry
Set_symmetry_colour_by_symop

IMAGES FROM PAUL EMSLEY

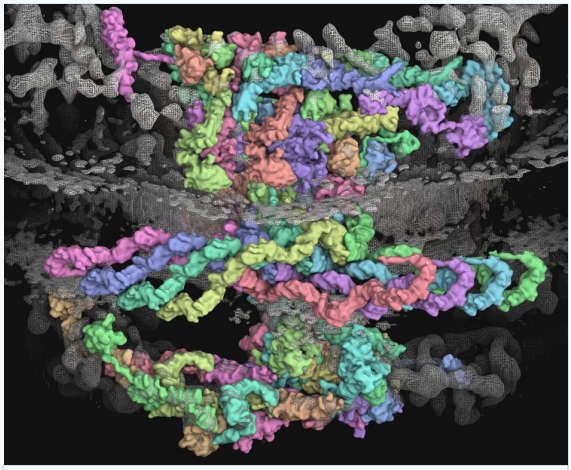
Coot 1.1 can produce Gaussian surfaces



Human Gut Microbiome Bacteriophage

IMAGES FROM PAUL EMSLEY

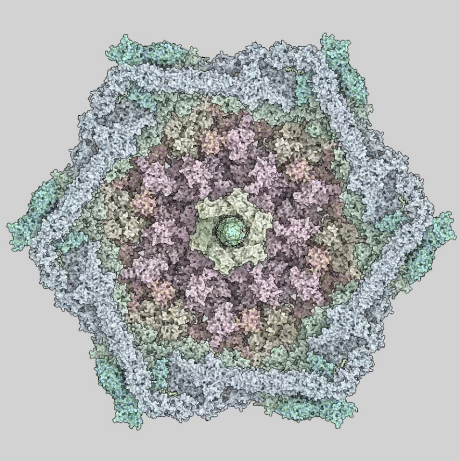
Coot 1.1 can produce Gaussian surfaces



Gaussian surface of Human nucleopore

IMAGES FROM PAUL EMSLEY

Coot 1.1 can produce molecular surfaces



Molecular surface of T4 baseplate

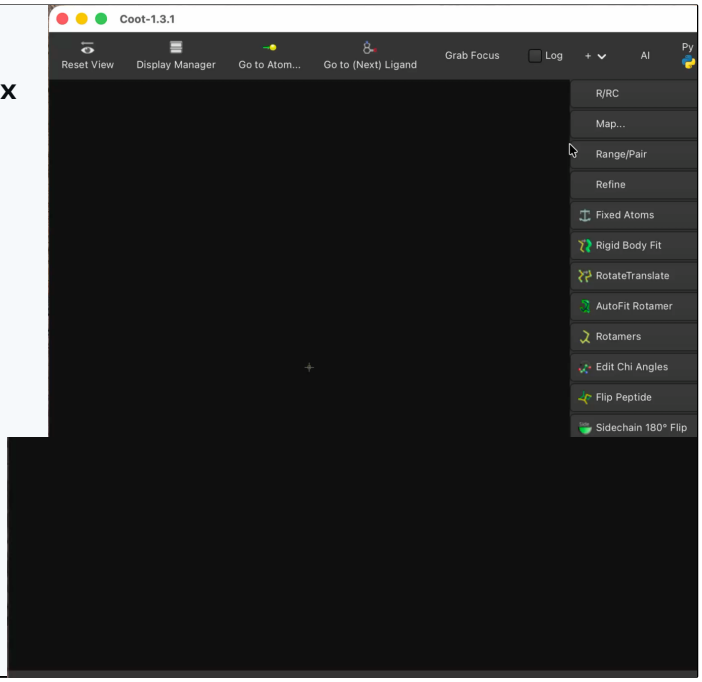
UNRELEASED INFORMATION

Coot 1.4 has PyMol-like syntax

Coot Commands

Set of commands which can control Coot, in a similar way to PyMol or ChimeraX

Can significantly speed up tasks



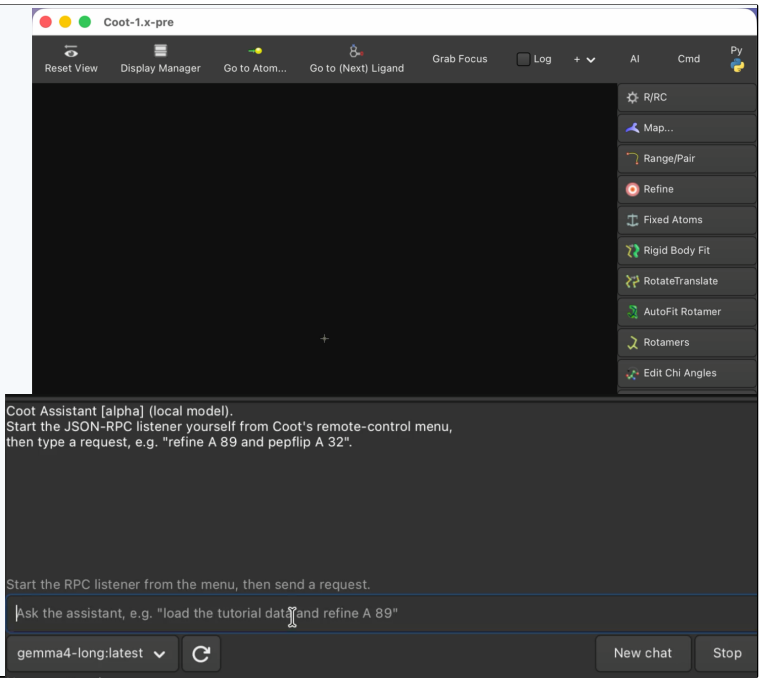
UNRELEASED INFORMATION

Coot 1.4 may have a local agentic assistant

Coot Assistant

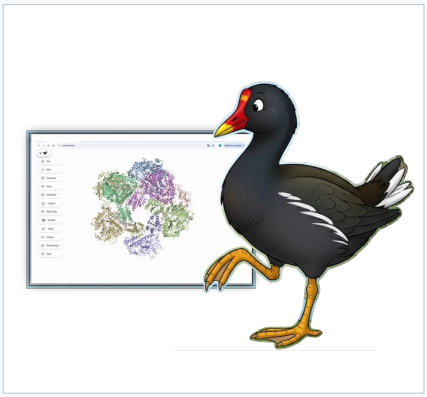
Uses a local large language model

Relies on the new Coot commands



Moorhen

Web-based interactive model building

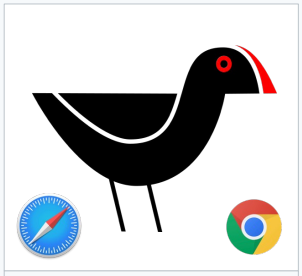


What is Moorhen?

Moorhen is a next-generation web-based application for visualization and manipulation of molecules for structure solution

In short, **Coot on the web browser**

Coot



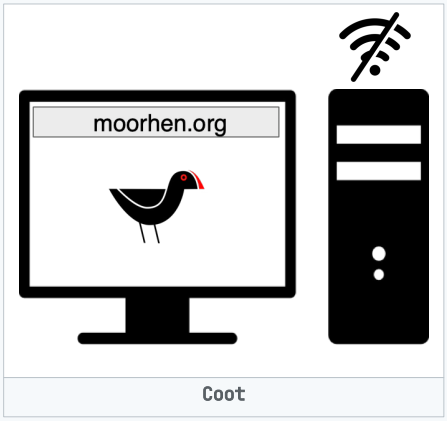
Moorhen

What is Moorhen?

Is it secure?

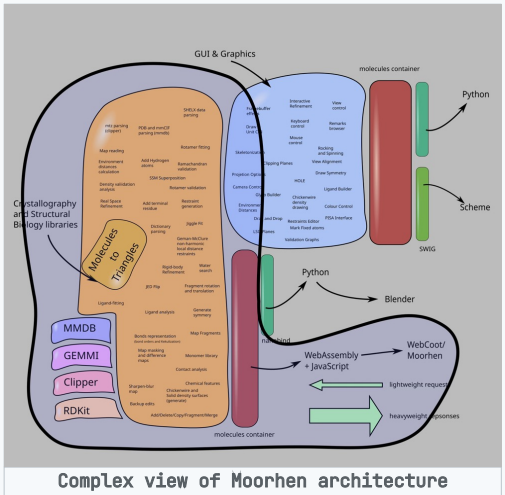
Moorhen is a **client-side-only** app

Once you open Moorhen, you can turn off your internet connection

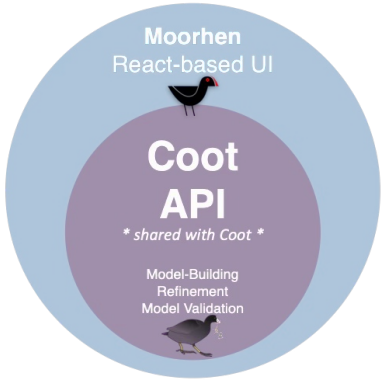


Coot

How does Moorhen work?



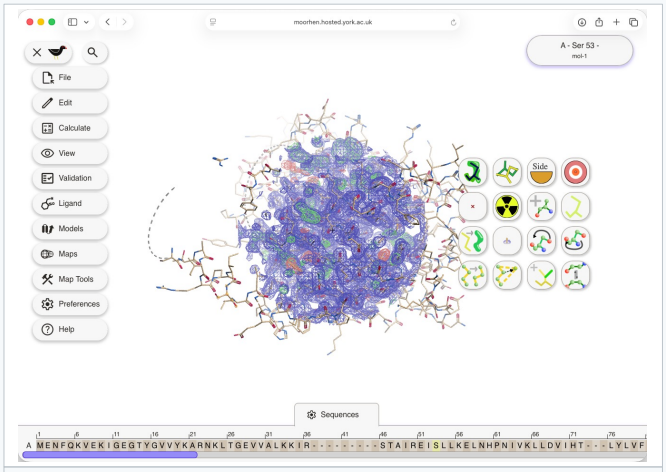
Complex view of Moorhen architecture



Simple view of Moorhen architecture

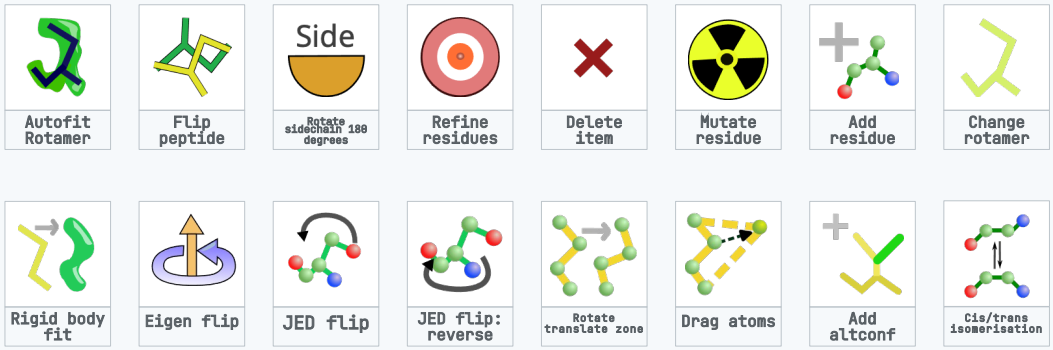
Moorhen is built on the same API that Coot uses, with a different interface.

What does Moorhen look like?



Moorhen in Safari

What tools are available in Moorhen for model editing?



Various tools available in Moorhen upon a right click.

What tools are available in Moorhen for validation?

Similar to Coot

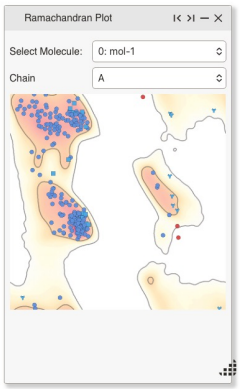
Density fit analysis

Geometry analysis

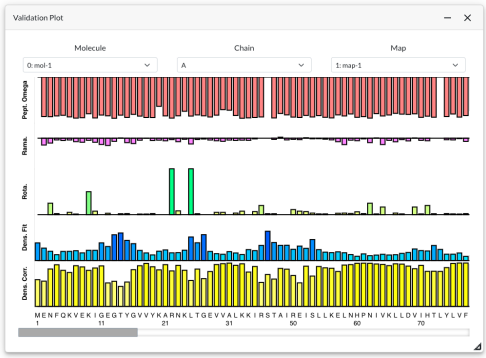
Rotamer analysis

Unmodelled blobs

Difference map peaks



Ramachandran plots



Validation plots

Interactive plots available in Moorhen.

What tools are available in Moorhen for figure making?

More than Coot

Multiple model representation styles

Arbitrary colour schemes

Shadows

Depth blur

Clipping and fogging

Ambient occlusion

Perspective projection

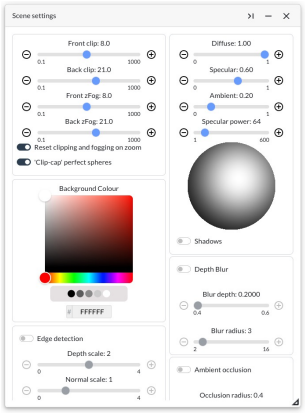
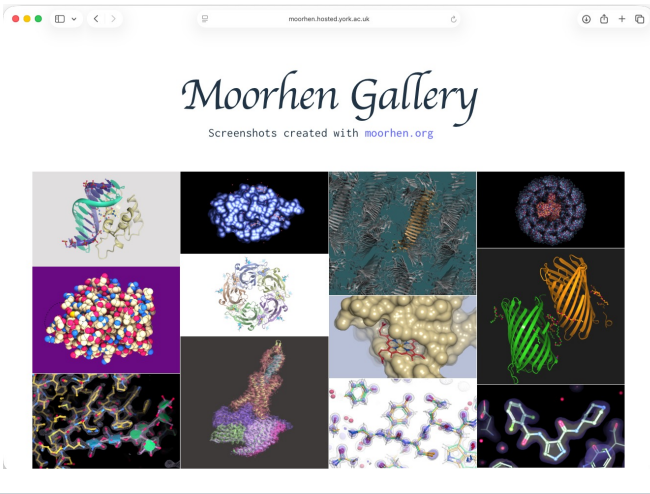


Figure making settings

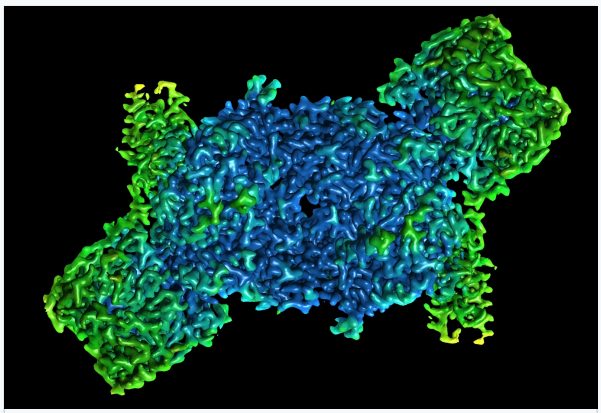
Where can I get inspiration?



Moorhen gallery contains some examples

IMAGES FROM LUCREZIA CATAPANO

Beautiful graphics from Moorhen

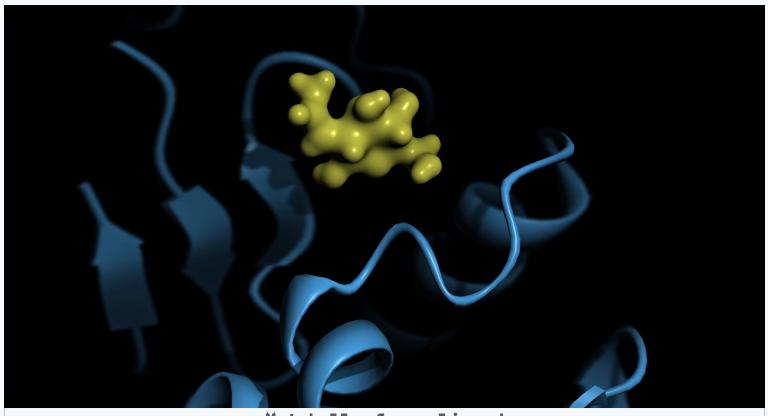


Occupy surface

Occupy tells you where particles in cryo-em reconstruction is incomplete

IMAGES FROM STUART MCNICHOLAS

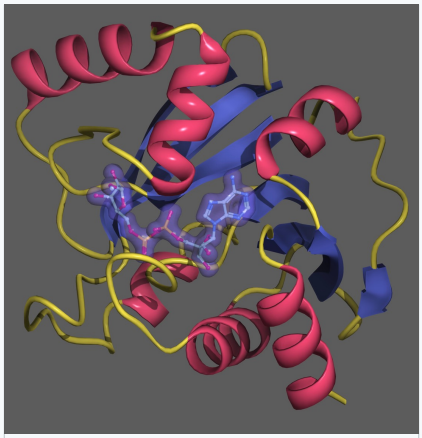
Beautiful graphics from Moorhen



Metaballs for a ligand

IMAGES FROM LUCREZIA CATAPANO

Beautiful graphics from Moorhen

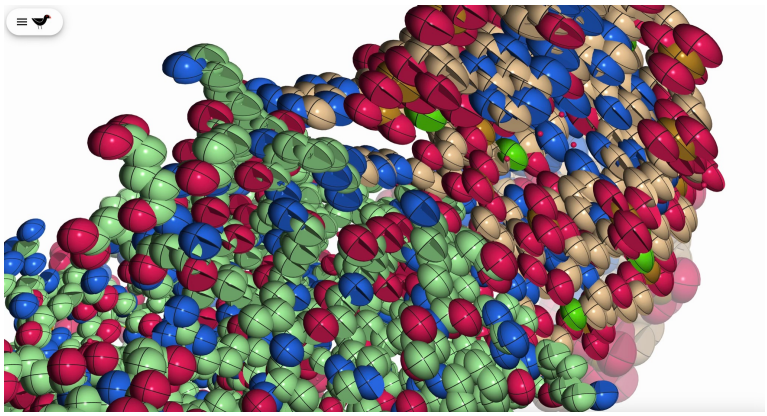


Colouring by secondary structure

Density selection and colour by secondary structure

IMAGES FROM LUCREZIA CATAPANO

Beautiful graphics from Moorhen

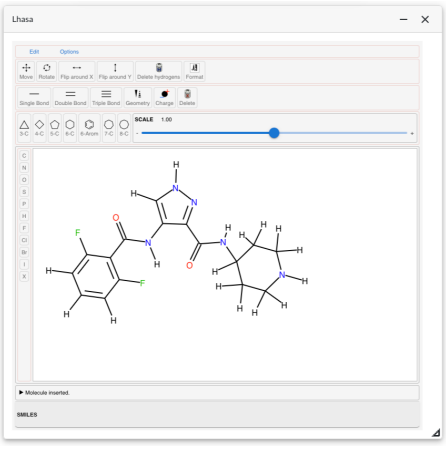


Thermal ellipsoids

Thermal elipsoids
Autoep

WORK BY JAKUB SMULKSI - GLOBAL PHASING LTD

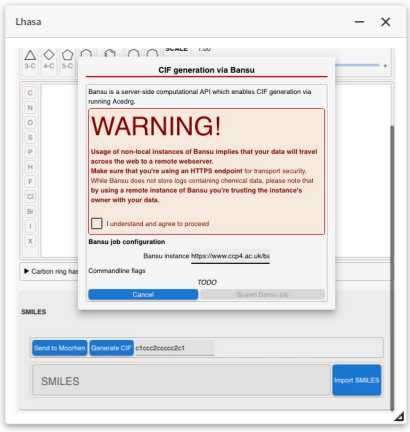
What ligand tools are in Moorhen?



Lhasa ligand builder in Moorhen

WORK BY JAKUB SMULKSI - GLOBAL PHASING LTD

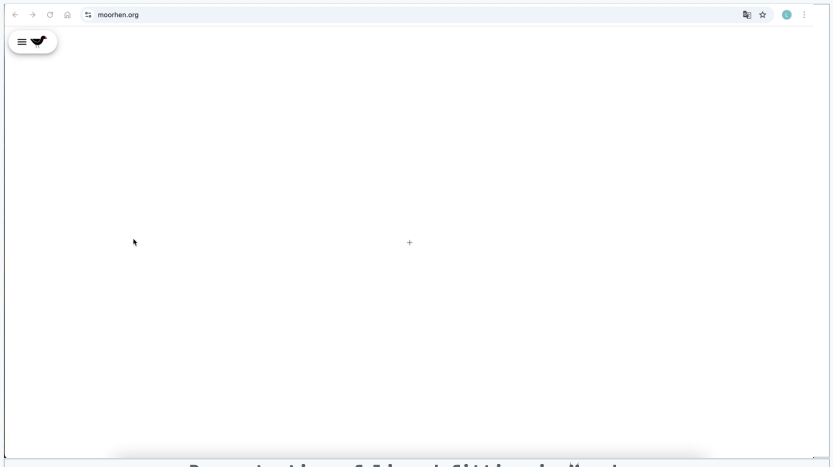
What ligand tools are in Moorhen?



Lhasa can generate ligand dictionaries

IMAGES FROM LUCREZIA CATAPANO

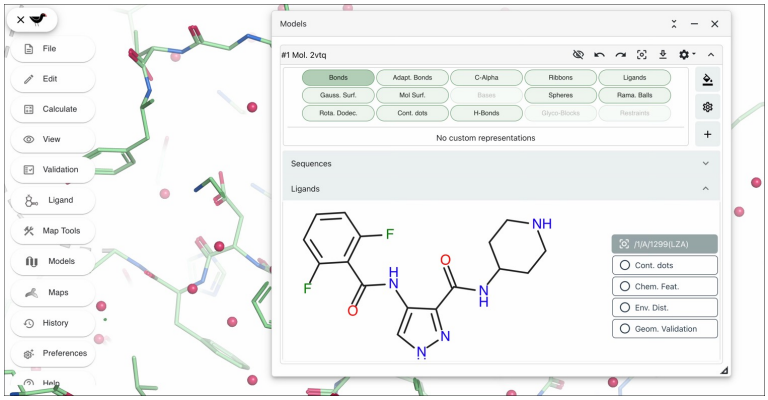
How can I fit my ligand in Moorhen?



Demonstration of ligand fitting in Moorhen

IMAGES FROM LUCREZIA CATAPANO

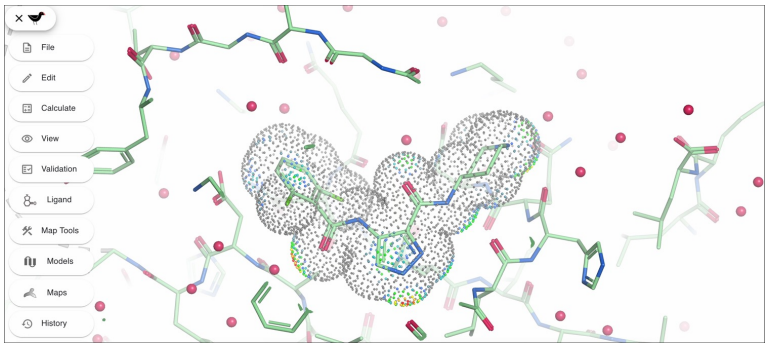
How can I validate my ligand in Moorhen?



View ligands in a model

IMAGES FROM LUCREZIA CATAPANO

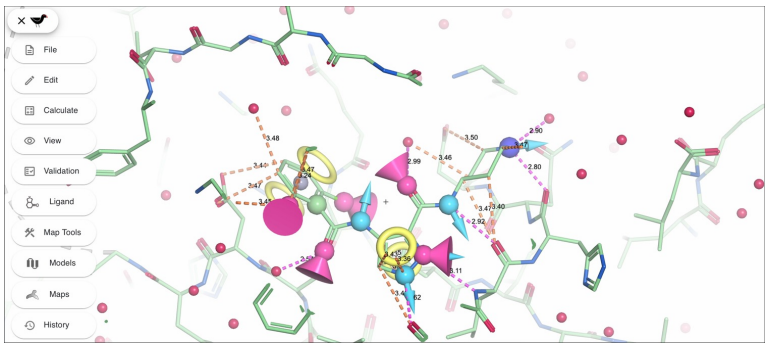
How can I validate my ligand in Moorhen?



Contact dots for ligand help show interactions

IMAGES FROM LUCREZIA CATAPANO

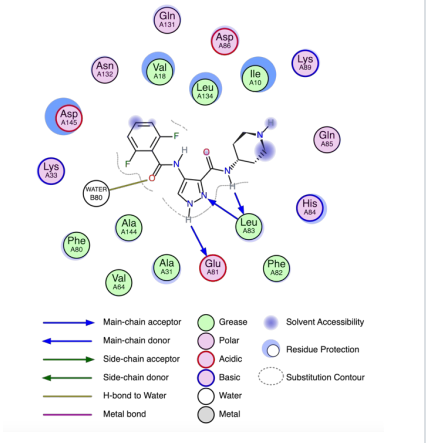
How can I validate my ligand in Moorhen?



Moorhen can show chemical features and environmental distances

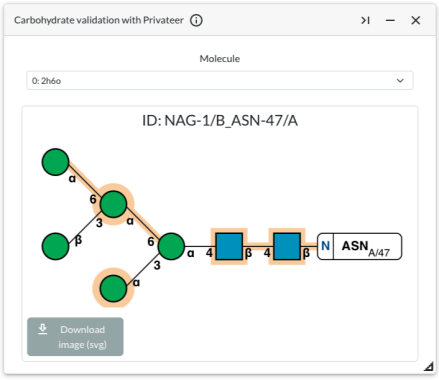
IMAGES FROM LUCREZIA CATAPANO

How can I validate my ligand in Moorhen?



Moorhen can show a 2D environment view

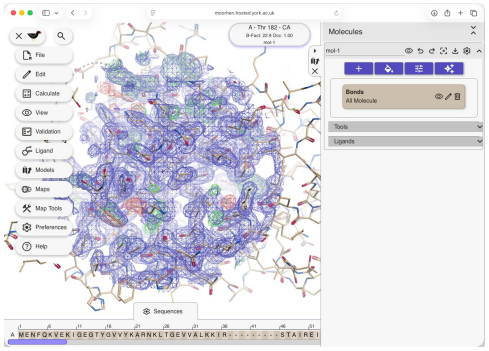
How can I validate my carbohydrates in Moorhen?



Moorhen uses Privateer to calculate carbohydrate diagrams

Where is Moorhen available?

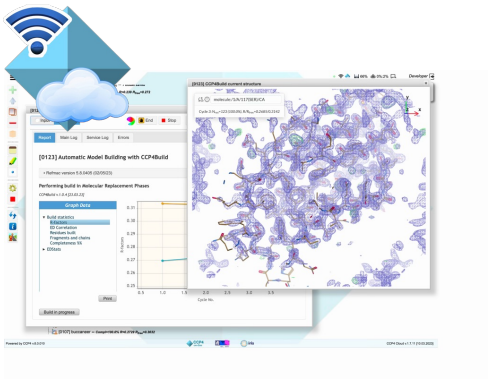
MOORHEN.ORG



Moorhen main website

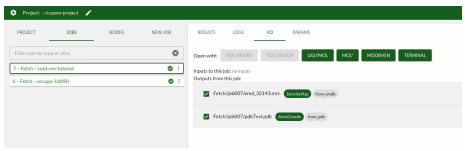
Where is Moorhen available?

CLOUD.CCP4.AC.UK



Available in CCP4 Cloud

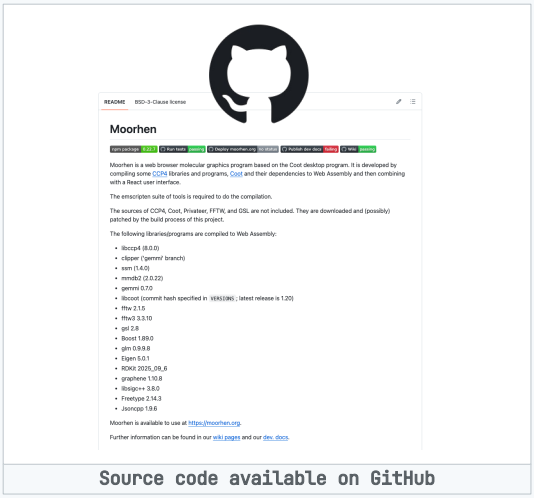
CCP-EM



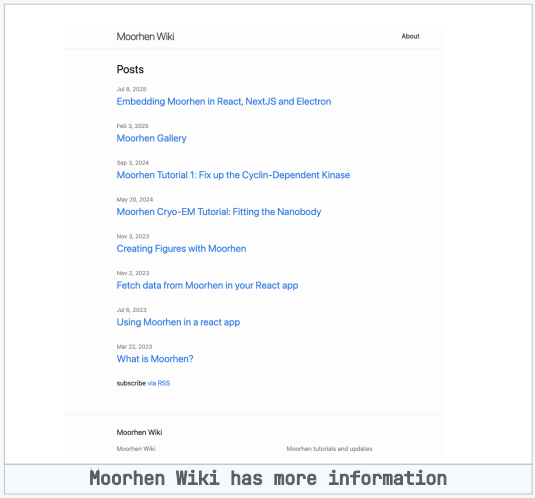
Available in CCP-EM Doppio

Moorhen is also inbuilt into CCP interfaces

What other resources are there for Moorhen?



Source code available on GitHub



Moorhen Wiki has more information

More information about Moorhen is available online.

Acknowledgements

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Martin Noble

Jakub Smulski

Bernard Lohkamp

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